Lab 3: Measuring Variability

Range, Interquartile Range, Variance, and Standard Deviation in SPSS

Deliverables: - SPSS data (`.sav`) file from SPSS. - An output (`.spv`) files from SPSS.
- A word or pdf document with answers to the take-home lab exercises at the end of the lab. The document must include images of your SPSS output tables and graphs where necessary.

PART A - Overview

In lecture and in the textbook (Chapter 4, *Variability*), we learned that variability describes how spread out or clustered together a set of scores is. In this lab, you will use **SPSS** to generate the four measures of variability covered in the chapter:

- the **range**
- the **interquartile range (IQR)**
- the **variance**
- the **standard deviation**

You will also learn about commonly used tools to visualize data variability of a continuous variable and how to report variability in APA style.

The Data

A psychologist is interested in understanding reaction time and how much reaction times vary from person to person. To investigate this, the psychologist asked 120 participants to complete a Stroop task, a commonly used task in psychology that measures how quickly people can respond when they encounter conflicting information. The variable Stroop_RT_ms records each participant's reaction time in milliseconds, with larger values indicating slower responses.

Download the dataset from this link then import it to SPSS. To import the data set, follow these steps:

File > Import Data > CSV Data..., browse to the file, and click **OK** on the import wizard (accept the default settings).

Make sure the measurement scales are set as follows:

- `Stroop_RT_ms` = Scale
- `Gender` = Nominal
- `Age` = Scale
- `ParticipantID` = Nominal

Why We Measure Variability

Recall from the reading: if every score in a distribution were identical, variability would be zero. In practice, scores differ from one another, and a good measure of variability tells us two things:

1. **How spread out the scores are** — are scores clustered close together or scattered widely?
2. **How well a single score represents the whole distribution** — the more variability there is, the less confidence we can place in any one statistic (or sample) as being representative of the group.

The textbook illustrates this with adult heights (low variability, tightly clustered around 70 inches) versus adult weights (much higher variability, spread widely around 170 pounds). Keep this contrast in mind.

> Checkpoint
>
> Without opening SPSS, which of the following would you expect to have **more** variability, and why?
>
> - The ages of students in a single 4th-grade classroom.
> - The ages of visitors at a public shopping mall.

Part B — The Range

The **range** is the distance between the largest and smallest score in a distribution:

$$\text{Range} = X_{max} - X_{min}$$

It is the simplest measure of variability, but it is also **unreliable** because it is based on only the two most extreme scores and ignores everything in between.

Get the Range in SPSS

1. Click **Analyze > Descriptive Statistics > Descriptives**.
2. Move `Stroop_RT_ms` into the **Variable(s)** box.
3. Click **Options...** and check **Range**, **Minimum**, and **Maximum**, then click **Continue**.
4. Click **OK**.

Look at the SPSS Output Viewer. You should see a table listing `N`, `Range`, `Minimum`, and `Maximum` for `Stroop_RT_ms`.

> Checkpoint
>
> What range does SPSS report for this data set? Confirm it by hand using the formula $\text{Range} = X_{max} - X_{min}$.

NOTE: Please DO NOT close the output window! You can just click on the main file showing the variables and data when you are ready to continue with the other analyses.

Part C — Quartiles and the Interquartile Range (IQR)

Because the range is so sensitive to extreme scores, researchers often prefer the **interquartile range**, which describes the spread of just the **middle 50%** of the distribution (between the 25th percentile, Q_1, and the 75th percentile, Q_3):

$$IQR = Q_3 - Q_1$$

The IQR trims off the top 25% and bottom 25% of scores, so a few unusually high or low scores do not distort it the way they distort the range.

Getting Quartiles in SPSS

There are two common ways to get the IQR in SPSS. Try both so you can see how they relate to one another.

Method A — Frequencies

1. **Analyze > Descriptive Statistics > Frequencies**.
2. Move `Stroop_RT_ms` into the **Variable(s)** box. (Uncheck "Display frequency tables" if you don't want the full table.)
3. Click **Statistics...**, and under **Percentile Values** check **Quartiles**. Click **Continue**, then **OK**.
4. SPSS reports the 25th, 50th, and 75th percentiles (Q_1, the median, and Q_3). Use the formula $IQR = Q_3 - Q_1$ to calculate the IQR.

Method B — Explore

1. **Analyze > Descriptive Statistics > Explore**.
2. Move `Stroop_RT_ms` into the **Dependent List**.
3. Under **Display**, choose **Statistics**. Click **OK**.
4. Look at the **Descriptives** table — SPSS directly labels a row **Interquartile Range**.

Checkpoint

1. What value did SPSS report for Q_1, Q_3 `Stroop_RT_ms`? What IQR value did you get by using the formula Range $= Q_1 - Q_3$?

2. Did Method A and Method B give you exactly the same IQR? (They may differ slightly — SPSS offers several algorithms for calculating percentiles. This is worth noticing, but don't worry about resolving the discrepancy for this lab.)
3. Compare the IQR to the *Range* for this same variable (`Stroop_RT_ms`). Which value is larger, and why does that make sense?

Part C — Standard Deviation and Variance

Standard deviation is perhaps the most important and most commonly used measure of variability. Conceptually, it measures the **standard (average) distance** between each score and the mean. The variance is the average of the *squared* deviations; the standard deviation is simply the square root of the variance, which brings the measure back into the original units of the data.

Both statistics are built from the **sum of squared deviations**, or SS. We obtain SS by subtracting the mean from each score, squaring the difference (deviation), the summing all squared deviations. Mathematically, this is abbreviated as;

$$SS = \Sigma(X - \bar{X})^2.$$

Population formulas (use μ for the mean and divide by N). The Greek letter σ^2 means population variance while σ means population standard deviation.

$$\sigma^2 = \frac{SS}{N} \qquad \sigma = \sqrt{\frac{SS}{N}}$$

Sample formulas (use M for the mean and divide by $n-1$ instead of n). The notation s^2 means sample variance while s means sample standard deviation.

$$s^2 = \frac{SS}{n-1} \qquad s = \sqrt{\frac{SS}{n-1}}$$

! Why n − 1? Degrees of Freedom

A sample mean is calculated *from* the sample, which places a restriction on the scores: once you know the mean and all but one score, the last score is no longer free to vary — it is fully determined. Only $n-1$ of the n scores are truly free to vary, so a sample has $df = n-1$ **degrees of freedom**. Dividing SS by $n-1$ (rather than n) slightly *increases* the resulting value, correcting for the fact that a sample's variability tends to underestimate the population's variability. This correction makes sample variance an accurate, **unbiased estimator** of the population variance.
This is exactly what SPSS's Descriptives procedure does by default — it always computes the *sample* formulas ($n-1$ in the denominator), because SPSS assumes your data are a sample unless you tell it otherwise. Keep this in mind whenever you are working with data that represent an entire population.

Standard Deviation for a Sample

We want to find the standard deviation (and other statistics) for the variable `Stroop_RT_ms`. By default, SPSS computes the sample standard deviation (i.e., divides the SS by $n-1$).

Follow these steps to find the sample standard deviation for the variable `Stroop_RT_ms`:

1. **Analyze > Descriptive Statistics > Descriptives**.
2. Move `Stroop_RT_ms` into the variables box, click **Options...**, and check **Mean**, **Variance** and **Std. deviation**. Click **Continue**, then **OK**.

Checkpoint

1. What sample mean (M), variance (s^2), and standard deviation (s) does SPSS report?
2. What does the standard deviation tell you about the spread of the `Stroop_RT_ms` scores?

Comparing Variability by Group (Sex)

Suppose we wanted to answer the question, "Do `Stroop_RT_ms` scores for females spread out more than `Stroop_RT_ms` scores for male students?

- Go to Analyze > Descriptive Statistics > Explore.
- Move `Stroop_RT_ms` into the 'Dependent List' box. Move `Sex` into the 'Factor List' box — this tells SPSS to split all output by male vs. female.
- Click the '**Statistics**' button and make sure '**Descriptives**' is checked (this gives you *variance*, *std deviation*, *min*, *max*, and *range* for each group). Select std deviation. Click '**continue**'.

Checkpoint

1. What sample standard deviation (s) does SPSS report for each group?
2. What is a possible explanation for the discrepancy in std deviations? (***Note***: there can be multiple correct answers here).

Part D — Visualizing Variability

Numbers alone may not always give you an intuitive feel for variability. Sometimes, graphs can be better tools to communicate the insights. **Box plots** and **histograms** are among the commonly used graphs to visualize variability of a continuous variable.

Box plots in SPSS

To create a boxplot for the variable `Stroop_RT_ms`, follow these steps:

- **Graphs > Chart Builder**
- From the gallery list, choose **Box plots**, then drag the **simple box plot** icon to the chart preview area.
- Next, drag `Stroop_RT_ms` to the y-axis.
- **OK**.

The box itself spans from Q_1 to Q_3 (the IQR), with a line in the middle representing the median. Whiskers extend to the most extreme scores that are not flagged as outliers. Any outliers are indicated with a dot or asterisk beyond the whiskers.

Histograms in SPSS

Create a histogram for `Stroop_RT_ms`. The steps are similar to those for a box plot. Let your instructor know if you need help with this. Once you have your histogram, you can add a normal curve overlay to help visualize how the distribution compares to a normal distribution.

Checkpoint

1. Describe the distribution of the `Stroop_RT_ms` scores. Be sure to talk about outliers, middle 50%, and the median.
2. Based on your histogram, how would you describe the spread of the `Stroop_RT_ms`?
3. How do box plots and histograms differ, and what specific information can be derived exclusively from each?

Part E — APA-Style Reporting

In journals following the APA style, the standard deviation (abbreviated as SD) and the mean (abbreviated as M) are reported in parentheses after the mean, e.g., $M = 12.45$, $SD = 3.70$. For skewed data, the median is reported with the IQR, e.g., $Mdn = 12.45$, $IQR = 3.70$.). Here is an example of a paragraph that reports the mean and standard deviation for a hypothetical memory recall test:

> Female respondents displayed higher memory recall ($\boldsymbol{M} = 12.45$, $\boldsymbol{SD} = 3.70$) than male respondents ($\boldsymbol{M} = 10.20$, $\boldsymbol{SD} = 2.80$).

"Checkpoint

Write one APA-style sentence summarizing the central tendency and variability of the `Stroop_RT_ms`. In your paragraph, compare the variability of `Stroop_RT_ms` for males and females.

Take-Home Lab Exercises

1. How spread out are the ages of the participants in this study? Use SPSS to compute any two different statistics to measure variability for the variable `Age`. Based on the two statistics, describe the variability of the ages in your own words.
2. Create a histogram and a box plot for the variable `Age`. Describe the distribution of ages, based on the box plot and the histogram.
3. How does the distribution of `Stroop_RT_ms` scores differ by gender? Create side-by-side box plots, with Gender on the x-axis, to compare the distributions of `Stroop_RT_ms` scores across genders. Describe your findings, including any differences in center, variability, and outliers.
4. Does the variability of `Stroop_RT_ms` differ by age group? Use SPSS to split the data into two groups: participants **younger than 25** and participants **25 or older**. Then, calculate statistics of your choice for each group that will help you compare the variability of `Stroop_RT_ms` scores. Describe your findings in APA style.

Note: Look up how to split the data in SPSS if you are not sure how to do this.

Submission Checklist

- ☐ A saved SPSS data file with correct value labels and measurement levels. Saving format: `Lab_3_Data_YourLastName.sav`.
- ☐ SPSS output containing all the analyses performed. Save format: `Lab_3_Output_YourLastName.spv`.
- ☐ Completed responses for the Questions above. This document should be typed (Times New Roman, Font size 12, double spaced). The document should include necessary SPSS output tables and graphs. Remember that SPSS output tables and graphs can be copied and pasted into a word document.